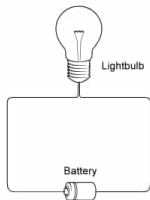


A battery is connected to a circuit joining it to a lightbulb as shown.

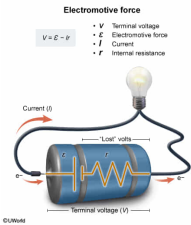


Increasing the internal resistance of the battery will:

- ☐ A. decrease the electromotive force. [29%]
- ☐ B. increase the electromotive force. [12%]
- ☒ C. decrease the terminal voltage observed. [33%]
- ☐ D. increase the terminal voltage observed. [25%]

Correct  
32% answered correctly

Explanation:



In electronics, the flow of charge (current) through a circuit requires the establishment of a gradient of **electric potential energy** (voltage) that is typically maintained by an **electrical generator**. Despite its name, the **electromotive force** ( $\mathcal{E}$ ) is the energy per unit of charge that is supplied by a battery or other generator to move charge through a circuit. In principle, the electromotive force represents an intrinsic electric potential gradient *within* the battery, which produces an observed **terminal voltage** ( $V$ ) that affects circuit components *external* to the battery (eg, resistors, conductors, capacitors). For an ideal battery, the observed terminal voltage is equal to the electromotive force:

$$V = \mathcal{E}$$

However, in real systems a battery uses chemical reactions to physically segregate positive and negative charges in a process that is not perfectly efficient. As a result, **internal resistance** ( $r$ ) within the battery dissipates some of the electric potential energy as charges are transported as current ( $I$ ) to the appropriate battery terminals. The voltage that is "lost" ( $V_{\text{loss}}$ ) due to internal resistance within the battery is expressed by application of **Ohm's law**:

$$V_{\text{loss}} = I r$$

Accordingly, the discrepancy between electromotive force and the observed terminal voltage is:

$$V = \mathcal{E} - V_{\text{loss}} = \mathcal{E} - I r$$

Consequently, the magnitude of the electromotive force is the theoretical upper limit of the terminal voltage, although this limit is not achieved in practical contexts.

Therefore, increasing the internal resistance of the battery connected to the circuit powering a light bulb will not change the magnitude of the intrinsic electromotive force; however, it will decrease the observed terminal voltage.

- (Choices A and B) The electromotive force is intrinsically related to the physical dimensions and composition of the battery; however, the magnitude of electromotive force is independent of internal resistance. Increasing internal resistance will not influence the magnitude of the electromotive force.
- (Choice D) For the observed terminal voltage to increase, the internal resistance of the battery would have to *decrease*, not increase.

Educational objective:

The electromotive force quantifies the capacity of electrical generators to establish an electric potential energy gradient. Due to internal resistance within the electrical generator, the observed terminal voltage is less than the electromotive force.

Time spent:0 sec  
Subject:  
Physics

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System:  
Electrostatics and Circuits